

10th International Conference of Information and Communication Technology (ICICT-2020)

Research and application of network status prediction based on BP neural network for intelligent production line

Yue Ma^{a,b}, Le Li^{a,b}, Zhenyu Yin^{a,b,*}, Anying Chai^{a,b}, Mingshi Li^{a,b}, Zhiying Bi^{a,b}

^aShenyang Institute of Computing Technology, Chinese Academy of Sciences, Shenyang 110168, China

^bUniversity of Chinese Academy of Sciences, Beijing 100049, China

Abstract

In the intelligent production line network communication process of the Industrial IoT, communication node congestion will cause the communication quality to decrease, thereby affecting the production efficiency. Therefore, accurately predicting the status of network and making adjustments to the network in real time is of great significance to improving the quality of network communication. Aiming at the urgent problem of the network communication quality of the intelligent production line, this paper proposes a network status prediction algorithm for the intelligent production line. The algorithm uses the ARMA prediction model to predict the network data, and calculates and predicts the entire network operation through the optimized BP neural network. At the same time, an intelligent production line network prediction system is designed based on the algorithm. The system can predict the network operation status in advance, reducing the impact of network status fluctuations on the production efficiency of the intelligent production line. The simulation results show that after a large number of network data prediction experiments, the optimal data prediction period is obtained. Under this period, the accuracy of network status prediction reaches 90%.

© 2021 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Peer-review under responsibility of the scientific committee of the 10th International Conference of Information and Communication Technology.

Keywords: ARMA; BP neural network; Network status prediction system

* Corresponding author. Tel.: +86-13017625798

E-mail address: 1074791723@qq.com

1. Introduction

In recent years, as the complexity of automatic production lines has increased, the possibility of failure of the entire network system has also increased. Generally, the reasons for network problems in automated production lines are: 1) the network data transmission rate is too high, causing network congestion, which in turn causes large delays, jitter, and packet loss; 2) the data transmission rate is too small, the current network bandwidth is not well utilized, and resources are wasted. The above two factors will have an impact on the operation of the production line. Moreover, due to the unpredictability of failures, relying only on manual experience to repair and inspect the network greatly reduces the work efficiency of the production line and wastes a lot of manpower and material resources.

Congestion or downtime of one or several network nodes in the production line network will cause the nodes to be unable to communicate, which will lead to the collapse or paralysis of the network. This will cause huge losses such as loss of transmission data and greatly reduce the work efficiency of the production line. Therefore, it is very important to predict network status such as network delays and congestion in advance so that the network operation and maintenance system can respond and deal with them in a timely manner, thereby ensuring the stable operation of the network. In order to avoid potential losses and damages caused by network problems, the automatic production line must be carefully inspected, and the status of the production line network at the next moment must be predicted, and the prediction result must be accurately given. With the rise of artificial intelligence technology, a large number of excellent prediction algorithms for big data analysis have emerged. The status prediction model established through the analysis of production line process data makes it possible to prejudge the operating status of production line equipment.

The existing network fault prediction methods mainly have analyzed the historical network fault data and use data mining methods to obtain fault prediction models. In [1], the author introduces state-based maintenance technology, and uses the constructed neural network to predict network failures by monitoring network status. In [2], established a multiple linear regression model for the failure rate of the network over a period of time, and then predicts the network failure rate in a period. In order to realize the self-management of failures and resources in the computer network system, a system with autonomous failure prediction and management capabilities is proposed in [3]. In [4] and [5], the authors proposed a multi-scale trend analysis of a single network parameter on the basis of frequent item sets mining of abnormal behaviors, and used multidimensional time-scale trends to calculate the network failures of nodes. In [6], the author proposed a prediction model for multi-zone faults, which uses risk assessment knowledge to describe the network fault model. Using neural network to predict the network status, domestic and foreign researchers have conducted a series of studies. In [11], the author aimed at the shortcoming that the basic predictive function control can only be used for the control of linear objects, and uses the combination of BP network and linear neural network to predict the nonlinear network traffic. In [12], this paper aimed at this non-linear network flow data, segmented the flow data according to time intervals, and applied the segmented data to the current linear neural network to predict the non-linear network data.

In the intelligent production line network communication process of the industrial IoT, communication node congestion will cause the communication quality to decrease, thereby affecting the production efficiency. Therefore, this paper proposes a network status prediction system based on the ARMA prediction model and optimized BP neural network algorithm. Firstly, this method predicts the status parameters of the network in the intelligent production line at the next moment, and then uses the predicted status parameters to accurately predict the network status of the intelligent production line.

2. Network Status Prediction System

The research idea in this paper is to use the network data of key nodes to predict the network status of the node. In this paper the network usage rate is used as an indicator. When the network usage rate reaches 60%-80%, we set it as the monitoring phase. More than 80% is defined as network failure. Less than 60% is normal operation¹⁰. Then calculate the overall network operation by monitoring the predicted network status of key nodes. The prediction system in this paper divides the intelligent production line network status prediction into three stages: data collection and preprocessing, network data prediction and network status prediction. It solves the problem of poor

communication quality due to the unpredictable increase in transmission delay of each transmission node in the intelligent production line, which in turn causes network failures such as communication interruption, and improves the production efficiency of the intelligent production line. The network status prediction process taken in this paper is shown in Fig. 1.

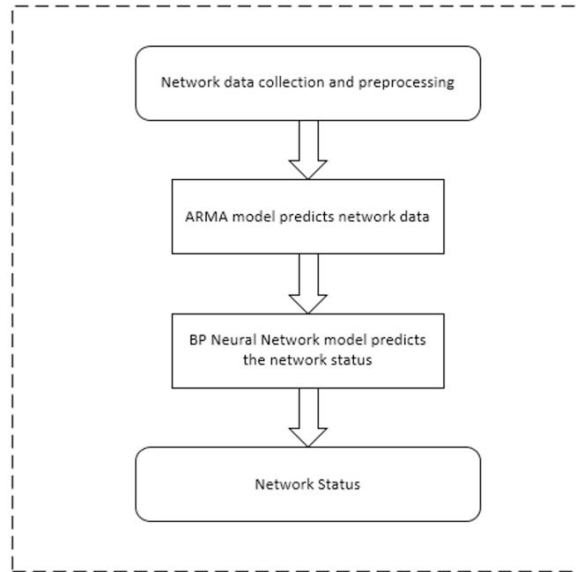


Fig. 1. The network status prediction process

2.1. Data Collection and Preprocessing

This paper mainly uses the SNMP (Simple Network Management Protocol) to collect network operating data in the intelligent production line. The SNMP is a very classic and commonly used network protocol. It has the advantages of simple principle and less system resources. It contains three parts: the subject of the agreement, the structure of management information, and the MIB (Management Information Base). Network flow data collection based on SNMP is an active collection method. The collection method is that the sender of the SNMP request and the server (switch, router, etc.) that respond to the request of SNMP cooperate to complete the collection. The SNMP's MIB library is shown in Fig. 2.

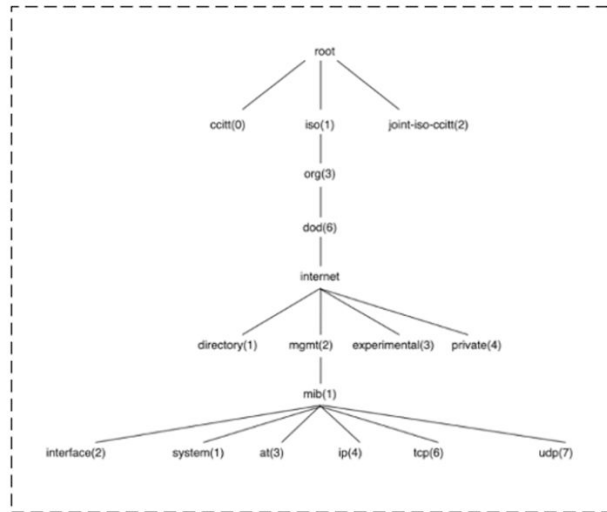


Fig. 2. SNMP 's MIB library

On the basis of analyzing and researching network failures, analyze and filter the attribute data of SNMP IP group, TCP group, Interface group and ICMP group to obtain the data required for network status prediction, and perform redundant data on the collected data preprocessing such as removing and cleaning up wrong data and data normalization for use in subsequent forecasting work. The collected data is mainly about network bandwidth usage rate, network packet loss rate and time delay. Use the collected network data for main research work. The main attributes collected in this article is shown in Table 1.

Table 1. Main attributes.

Attribute	Collection Method	Data Type
ipDefaultTTL	walk	Integer
ipOutDiscards	walk	Counter
ipOutnoRoutes	walk	Counter
ipInDiscards	walk	Counter
ifInOctets	walk	Counter
ifSpeed	walk	Gauge
ifInDiscards	walk	Counter
ifOutDiscards	walk	Counter
ifOutOctets	walk	Counter
ifInOctets	walk	Counter
tcpRtoMin	walk	Counter
tcpRtoMax	walk	Counter
tcpRetransSegs	walk	Counter

In this paper, data cleaning, redundant attribute data removal, data normalization processing and other pre-processing operations were carried out before data use. The data cleaning mainly deletes some error data and some irrelevant header and tail data, and normalizes the data. The normalization processing of data is mainly to aggregate the data of each node and combine the data of each property to form a available data set. Sort and classify the collected attribute data, and remove and clean up the head and tail data that is not important to the network. Because

there are some normal data in the beginning and end of the collection phase when collecting blocked data, and when the fault data is generated, it is an accumulation process. It belongs to the normal status before it is accumulated to the fault status. The data cannot be used as the fault data for calculation and processing at this time. Therefore, such data must be deleted. There is also a failure of the monitoring node during the collection process, which makes the collected data become abnormal. For example, some traffic data suddenly becomes negative, and these data need to be cleaned up. After the above cleaning, the collected network operating data needs to be preprocessed so that the processed data meets the applicable conditions of the sliding autoregressive model.

2.2. Network Data Prediction

This paper firstly uses the collected network operating data of the intelligent production line to establish a network status parameter prediction model, and uses the network status parameter prediction model to predict the network operating parameters at the next moment. The network status parameter prediction model is divided into two parts: 1) preprocess the collected intelligent production line network operating status data into a kind of data that meets the conditions of the sliding autoregressive model; 2) use the ARMA model for data prediction and get good results. The network status parameters at the next moment. According to time series theory, the model ARMA(2n, 2n-1) (n is a positive integer) is usually selected. This article predicts network traffic and other parameters, and the change of traffic is in a trend and cannot be predicted with a small amount of data, so the form of ARMA(6,5) is chosen, which can be expressed as formula (1).

$$y_t = \sum_{i=1}^6 \varphi_i y_{t-i} - \sum_{j=1}^5 \theta_j \varepsilon_{t-j} + \varepsilon_t \quad (1)$$

2.3. Intelligent Production Line Network Status Prediction

The predicted network operating parameters data are input into the BP neural network model for prediction, and the three-layer structure BP neural network is trained and learned using the sample data of the network operating status. This paper selects the five attributes of interface receiving data, interface sending data, TCP connection reset number, memory usage, and CPU usage as input parameters of BP neural network. Suppose the number of input layer nodes of the BP network is n, the number of output layer nodes is 1, the hidden layer is a layer structure and the number of nodes is m; the input layer node of BP corresponds to the network operating status parameters collected at a certain time, and the output layer node represents the current network status (network usage rate). In this paper, the hidden layer and output layer of the BP neural network model are calculated using the nonlinear function of formula (2). The network status at the next moment is predicted according to the input network operating status parameters, and the network operating status at the next moment is obtained. Realize the prediction of the network status of the intelligent production line.

$$o_j = \frac{1}{1 + \exp(-\sum_{i=1}^k w_{ij} x_i)} \quad (2)$$

The w_{ij} in formula (2) is the connection weight between the i-th node of the previous layer and the j-th node of this layer. When o_j represents the output of the intermediate layer, $1 \leq j \leq m$, $k=n$; when o_j represents the output of output layer, $j=1$ and $k=m$.

In this paper, the random number in the interval (-1,1) is set as the initial value of w_{ij} . The learning of the weight is adjusted according to the error between the output value of the output layer node and the output expected value. The weight w_{ij} of the BP network uses the formula (3) to make adjustments.

$$w_{ij}(n+1) = w_{ij}(n) + \eta o_i \delta_j \quad (3)$$

In formula (3), n and n+1 represent the number of generations, η is the learning factor which is set to a constant (0.8 in the paper), o_j is the output of the i-th node of the previous layer (if the i-th node is in the input layer, then the

output value is the input value), if j is the output layer node, $\delta_z = o_z(1 - o_z)(t - o_z)$, if j is the hidden layer node, $\delta_j = o_j(1 - o_j)\delta_z w_{jz}$.
The main algorithm process is shown in Table 2.

Table 2. The main algorithm.

Algorithm	Network status prediction Algorithm
Input:	
DataSet // Dataset collected by SNMP	
Output:	
1: Begin	
2: DataSet = SNMP MONITOR // Put the collected data into the dataset	
3: Foreach (DataSet) // Iterate each dataset	
4: {	
5: PreData = ARMA(DataSet(i)); // Predict the i-th node data using ARMA	
6: Network Status = BP(PreData); //Predict network status using BP neural network	
7: }	
8: End	

3. Simulation Experimental Analysis

In this paper, a simulation platform based on Matlab, Simulink and Truetime toolbox is used to build a simulation platform for intelligent production line network status prediction.

3.1. Network data prediction experiment

The network data prediction experiment is to use the network data which collected by SNMP, after data preprocessing, input into the ARMA parameter prediction model after training and learning to predict the corresponding parameter values of 5min, 10min, and 15min in the future. The Fig. 3 shows three part of the Ifinoctets prediction results of the network operating parameters.

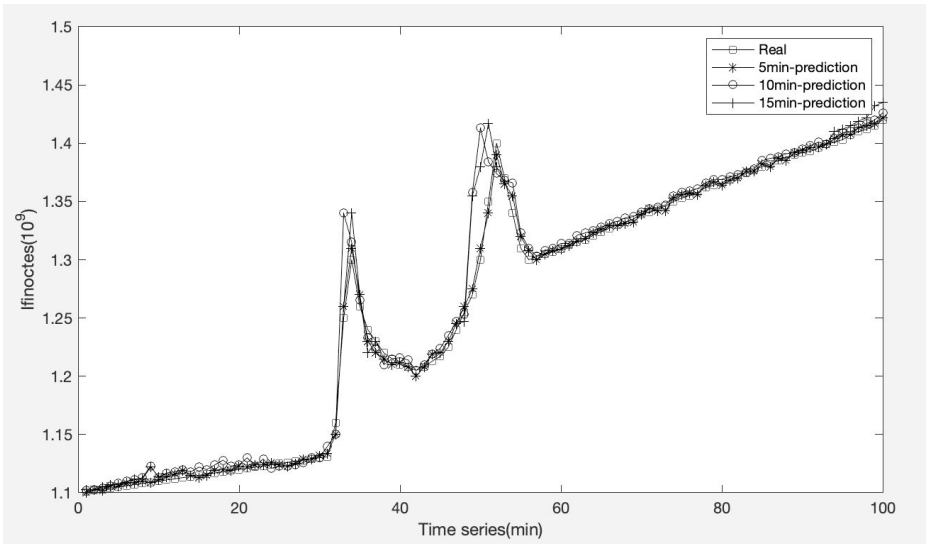


Fig. 3. Ifinoctets prediction

It can be seen from the comparison chart of the results of parameter prediction and actual measurement that the predicted parameter can get the change trend of the parameter in advance, but the calculated result of the predicted value to the change value of the parameter is somewhat too large. The relative error of parameter prediction can be calculated. If the predicted value of a certain parameter is a and the measured value is b , then the relative error of parameter prediction is shown in formula (4).

$$e = \frac{|a-b|}{b} \quad (4)$$

Table 3 shows the average relative error of the predicted value of three network operating parameters for 5 minutes, 10 minutes and 15 minutes. It can be seen that the farther the forecast time is, the greater the forecast error will be. It is reasonable.

Table 3. Mean relative error

Mean relative error	5min	10min	15min
Ifinocets	0.267	0.984	2.695
Ifoutocets	0.119	0.472	0.963
Tcpstabresets	0.591	1.587	3.764

3.2. Network status prediction experiment

The experiment has tested the accuracy of the improved BP neural network in predicting the status of the network. In the experiment, in order to maintain the working status of the network, the data transmission program is run at the monitoring points of the network, so that the network utilization rate is about 30%. The network operation parameters are collected at the network monitoring points, and then the network operation parameters are measured. After 5 minutes of numerical prediction, the predicted value is input into the trained BP neural network, and the result of the network running status after 5 minutes is obtained. The experimental results are shown in Fig. 4.

Because the network operating parameter prediction model is sensitive to the change prediction of network operating parameters, and the network operating status prediction is based on the predicted network operating parameters, it shows that the network operating status prediction can predict the changes in network operation in time, but the prediction The magnitude of change is larger than the real magnitude of change.

Assuming that the absolute value of (predicted network utilization rate-actual network utilization rate) is less than 2%, the prediction is considered to be basically accurate. The test in Figure 4 is greater than 90%. Basically, the predicted goal can be achieved.

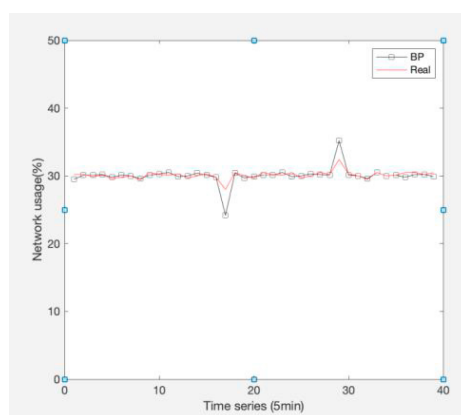


Fig. 4. Network status prediction.

4. Conclusion

Aiming at the problems that need to be solved urgently in the communication quality of the intelligent production line network of the industrial Internet of IOT, this paper carries out the research of the intelligent production line network status prediction system based on the optimized BP neural network, and pre-judges the operation status of the intelligent production line network, reducing the fluctuation of the network status. In this paper, the intelligent production line network status prediction is divided into three stages: data acquisition and preprocessing, network data prediction and network status prediction. Firstly, the SNMP protocol is used to collect the network operation data in the intelligent production line, and then the network status parameter prediction model is established to predict the network operation parameters at the next time by using the network status parameter prediction model, and finally input into the optimized BP neural network model for prediction. The system solves the problem of poor communication quality due to the unpredictable increase in transmission delay of each transmission node in the intelligent production line, which in turn causes network failures such as communication interruption, and improves the production efficiency of the intelligent production line. The simulation results show that after a large number of network data prediction experiments, the optimal data prediction period is obtained. Under this period, the accuracy of network status prediction reaches 90%.

Acknowledgment

This work is supported by LiaoNing Revitalization Talents Program under Grant NO. XLYC1802112.

References

1. Xiaokai Hou, Shiqian Li, Jieqiong Wang, et al. A neural network-based network equipment fault prediction system[J]. Journal of Shandong University of Technology: Natural Science Edition, 2014, Issue 6: 29-34.
2. Deng Li, Fan Geng, Liu Zhixue. Network fault prediction based on regression analysis method[J]. Computer Engineering, 2012, 38(20): 251-255.
3. Zhang Z, Fu S. Failure prediction for autonomic management of networked computer systems with availability assurance. [C]// 2013 IEEE International Symposium on Parallel Distributed Processing, Workshops and Phd Forum IEEE, 2010: 1-8.
4. Abed H, Al-Fugaha A, Guizani M, et al. Failure Prediction Based on Multi-Scale Frequent Anomalous Behavior Identification in Support of Autonomic Networks[c]// Global Telecommunications Conference (GLOBECOM 2010), 2010 IEEE, 2010: 1-5.
5. Abed H, Al-Fugaha A, Khan B, et al. Efficient failure prediction in autonomic networks based on trend and frequency analysis of anomalous patterns. International Journal of Network Management, 2013, 23(3): 186-213.
6. Liu Xiao, Wang Ying, Xiao Ailing, et al. Disaster-prediction based virtual network mapping against multiple regional failures[c]// Integrated Network Management (IM2015 IFIP/IEEE International Symposium on IEEE, 2015.
7. Huang J D. A Study of Equipment Fault Forecast System Based on Virtual Setting[J]. Applied Mechanics & Materials, 2011, 80-81:1244-1248.
8. Fanxue Meng, Yanheng Liu, Jing Wu, et al. Network traffic analysis based on distributed statistical time series[J]. Computer Science, 2010, 37(07): 110-114.
9. Hongyu Zhou, Jian Zhang, You Like, Nonlinear Predictive Function Control Based on Hybrid Neural Network[J]. Control Theory and Application: 2005, 22(1):110-113.
10. Yue Ma, Anying Chai, Bihui Yu, Bo Yu. A dynamic control technology for multi-party calls [J]. Computer System Applications, 2018, 27(03): 64-70.
11. Jie Liu, Yalou Huang. Non-linear network traffic forecast based on BP neural network[J]. Computer Applications, 2007, 27, (7):1770- 1772.
12. Yang J, Ting C H, Zhang X D. Network Traffic Prediction Using the Combination of Chaos Theory and Simple-MKL[J]. Journal of Networks, 2013, 8 (8).
13. Zhongda Tian, Shuijiang Li, Yanhong Wang, et al. Application of Empirical Mode Decomposition and Time Series Analysis in Network Traffic Forecasting[J]. Control and Decision, 2015, 30(5):905-9.
14. Liang Liu, Hanhong Jiang, Jie Wang, et al. ARMA-SVIR network traffic prediction method based on wavelet analysis[J]. Computer Engineering and Design, 2015(8):2021-2025.
15. Ping Chen, Yuren Wei. Research on status prediction method of electronic equipment based on GRNN neural network[J]. Ship Electronic Engineering, 2019, 39(01):98-100+126.
16. Qicai Zhou, Yifei Wang, Jiong Zhao, Xiaolei Xiong, Hehong Shen. Shield machine fault prediction system design based on LSTM recurrent neural network[J]. Modern Machinery, 2018(05):35-40.
17. Jian Zhou, Genlin Zhao, Youwei Huang. Multi-node fusion decision concurrent fault diagnosis based on hidden Markov model[J]. Application Research of Computers, 2015, 32(8):2447-2449.
18. Omidy Oly, Shujie Liu. Overview of the Importance of Intelligent Approaches on Machinery Faults Diagnosis and Prediction Based on Prognostic and Health Management/Condition -Based Maintenance[J]. Journal of Donghua University (English Edition), 2018, 35(03):270-273.